Kewei Pan

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Research interests: Random matrix theory, noncommutative probability, entropy and functional inequalities, large deviations.

Education

Université de Toulouse *Oct. 2023 – Oct. 2026*
Ph.D. in Mathematics
Thesis: Boolean analogues of Shannon's entropy in one dimension: microstates and non-microstates approaches.
Advisor: Guillaume Cébron; co-advisor: Djalil Chafaï (École normale supérieure de Paris)

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Université de Toulouse *Sept. 2022 – July 2023*
M.Sc. in Mathematics
Selected coursework and grades: Convergence of Probability Measures (19/20), Stochastic Calculus (20/20), and Random Matrices and Free Probability (19.5/20).

University of Science and Technology of China (USTC) *Sept. 2018 – June 2022*
B.Sc. in Mathematics
Weighted average: 86.82/100; major GPA: 3.80/4.30; overall GPA: 3.60/4.30.

List of Publications

- K. Pan, A Boolean analogue of Shannon's entropy and the monotonicity problem, *Advances in Mathematics*, vol. 503 (2026).

Preprints

- K. Pan, Entropic inequalities and limit theorems in Boolean probability, arXiv:2610.20178 (2026).
- K. Pan and G. Cébron, Functional inequalities for Boolean entropy, arXiv:2602.23527 (2026).

Selected Talks and Seminars

- Workshop on Entropy and Random Matrices, CIRM, Marseille, France — Mar. 2026
- Voiculescu's Probabilistic Operator Algebras Seminar (online) — Mar. 2026
- GDR MEGA Autumn School, Les Plantiers, France — Oct. 2025
- Journée de rentrée de l'équipe Probabilités, Institut de Mathématiques de Toulouse (IMT), Toulouse — Oct. 2024

Teaching Experience

Teaching Assistant, Linear Algebra, University of Science and Technology of China (USTC) — Spring 2022

Awards and Funding

LabEx CIMI and EUR MINT funding — Oct. 2023
CIMI Fellowship for the M2 programme — Sept. 2022
USTC Endeavour Award — Fall 2021
USTC Excellent Student Award — Fall 2018, 2020, 2021
Second Prize, Chinese Mathematics Competition (CMC) — Fall 2019

Skills and Languages

Programming: Python, C/C++, R, MATLAB. **Scientific software:** LaTeX, Mathematica.
Languages: Chinese (native), English (fluent), French (B1, intermediate).